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LATA: A Pilot Study on LLM-Assisted *Thematic Analysis* of Online Social Network Data Generation Experiences

Oct 21, 2025

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Protocol # 2037012-1

Thematic Analysis (TA)

Systematic **Qualitative** Research Method

Datasets such as interviews or written materials

Identify, **Analyze**, & **Report** Patterns

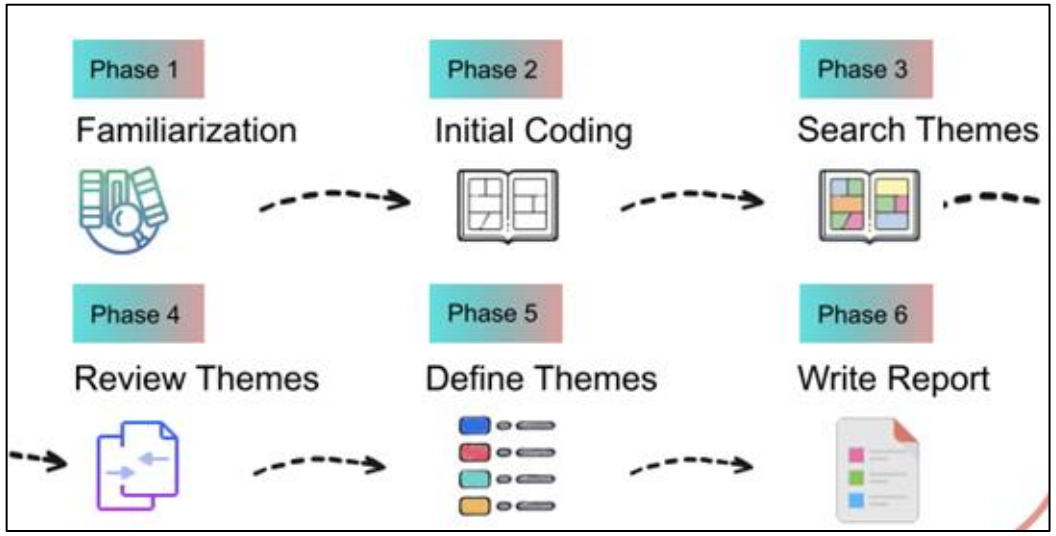
Introduced by Braun & Clarke



Thematic Analysis (TA)

Challenges :

- **Time-consuming** and Manual work
- Require specific software, analytic skills
- At least 2 researchers
- Maybe inconsistency between different group of researchers

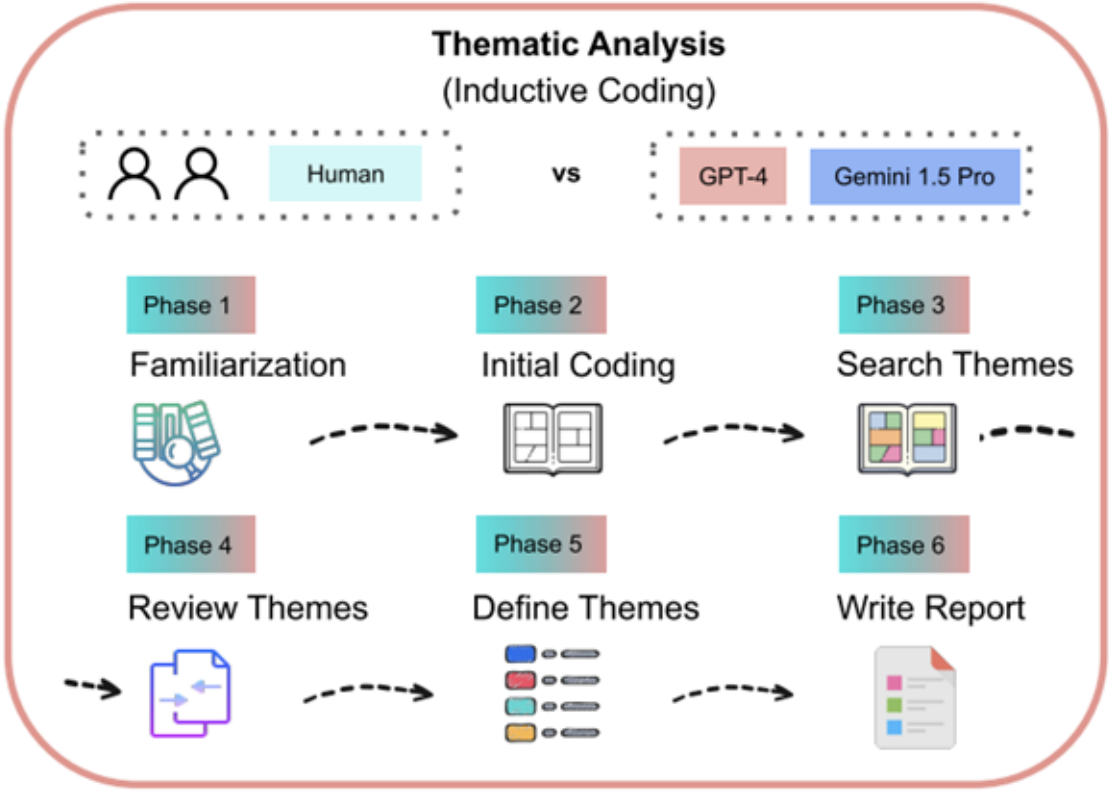


Frameworks - 6 Steps (Braun & Clarke, 2006)

Motivation

Given **limitations of manual thematic analysis(TA)** and the popularity of LLM, we are motivated to explore:

- *RQ1: Can LLM assist in performing TA?*
- *RQ2: How do the outcomes of TA using LLMs compare to those obtained via human experts?*



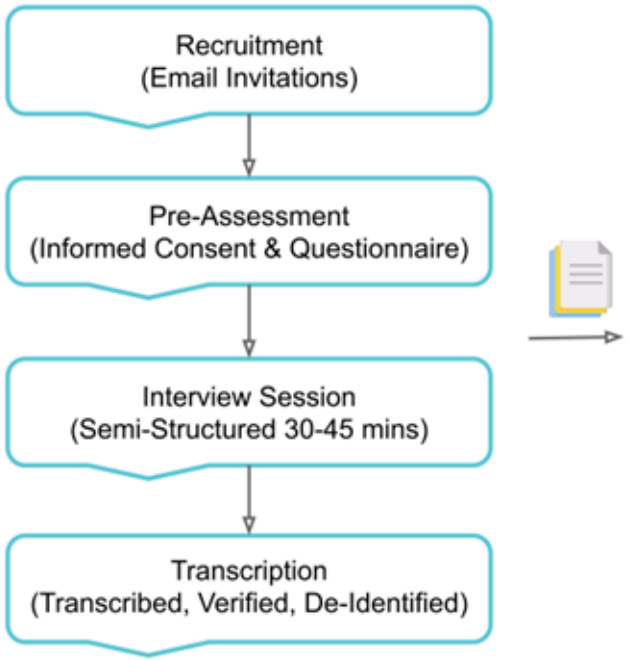
Interview Study - OSN

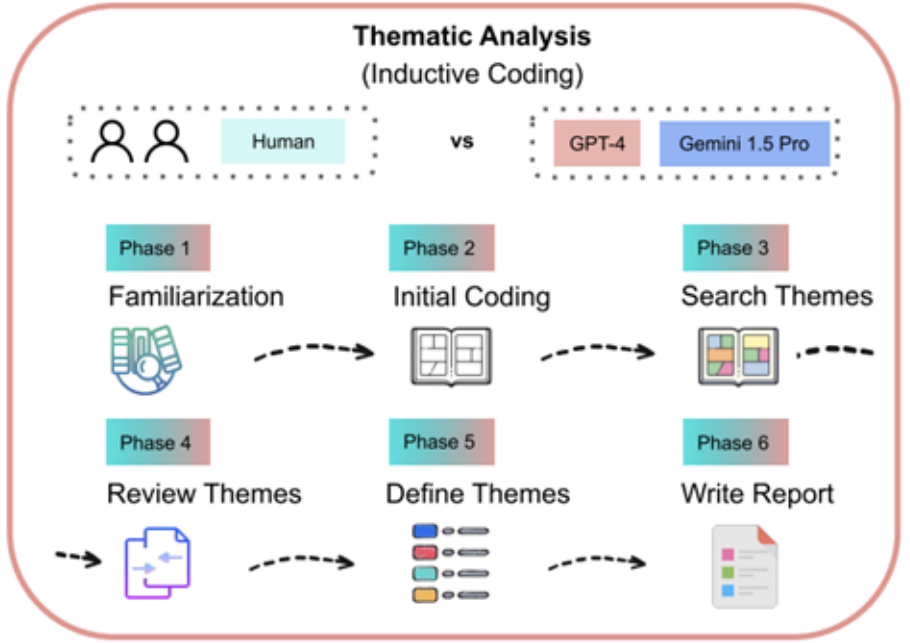


Interview data is generally not available, so we conducted a semi-structured interview study

Topic Selected:

- Online Social Network (OSN)
- *RQ: We are interested to learn how researchers experience working with OSN and how can it be enhanced?*





Participants

N=14 Researchers

Age: 28 – 64 (41.4 ±11.3)

57% Male

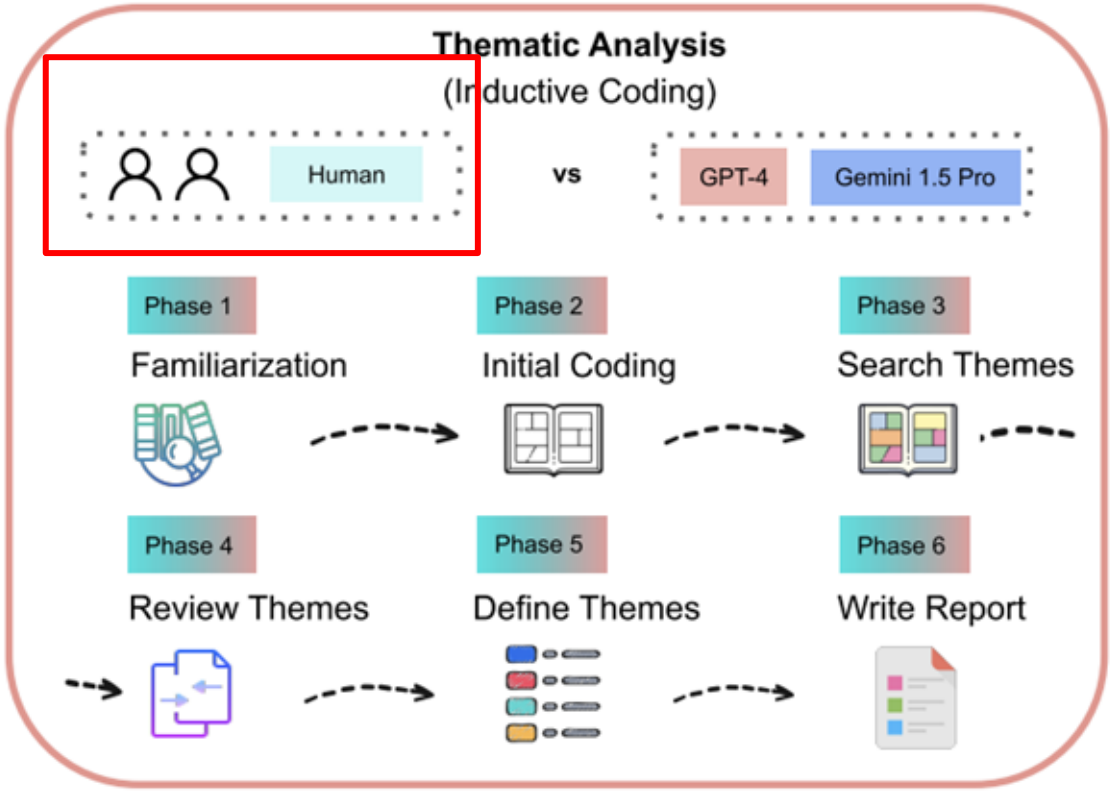
78.57% 5+ yrs of Research Experience



Primary Research Domain

Method

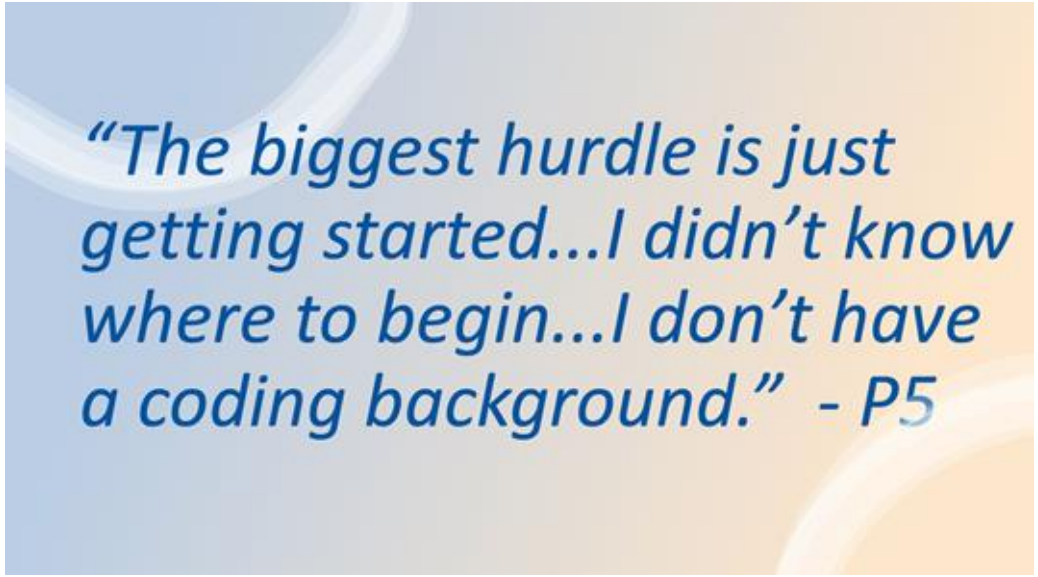
- Human Expert Analysis (Manually)
  - Two researchers
  - via Dedoose (software)
  - Developed coding directly from the data (Inductive Coding)



Insights from Interview Study - OSN

Learning recent OSN research experience

- Noisy Data
- Technical Skill Requirement
- Poor or Missing API Documentation
- Manual Work (not scalable)
- Ethical Issues (e.g. use deleted historical posts)



Insights from Interview Study - OSN

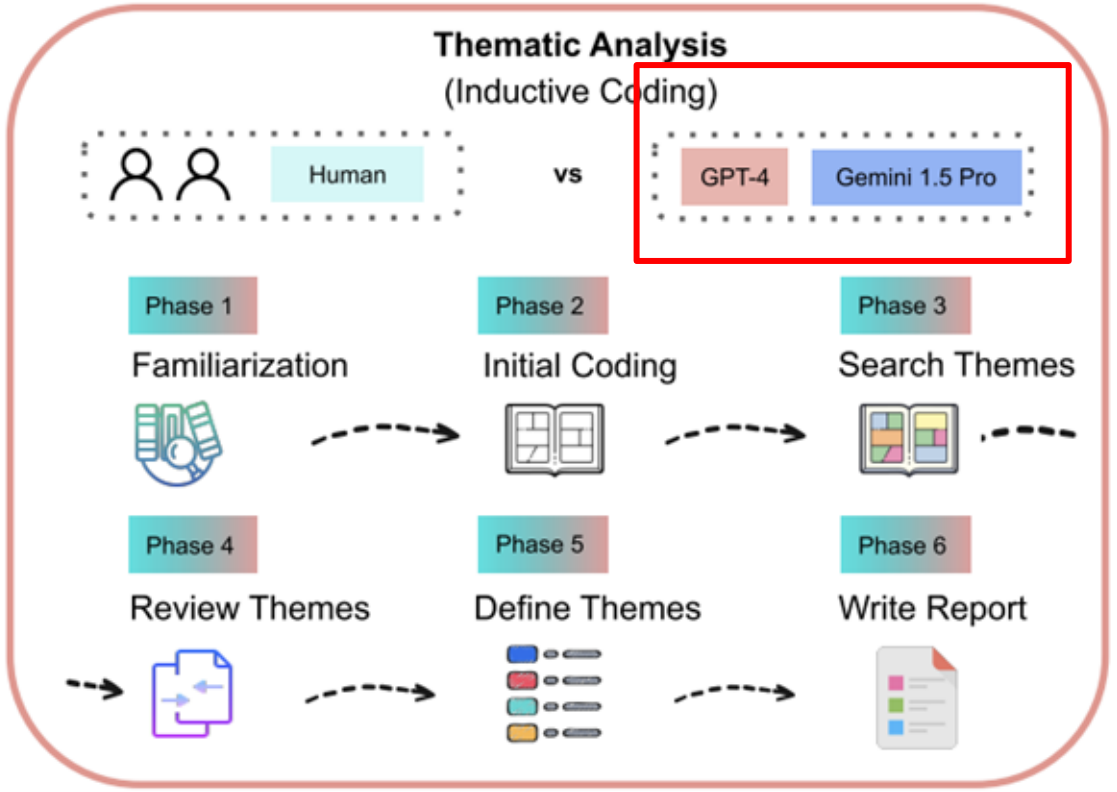
Participants prefer (in order):

- Historical data
- User-friendly interface & navigation
- Advanced search filters and queries
- Real-time data monitoring and collection

Method

- Human Expert Analysis (Manually)
- **LLM-based Analysis**
  - GPT 4 & Gemini 1.5 Pro
  - No Codebook provided (inductive coding)
  - Parameter Temperature = 0 to ensure reproducibility
  - Intermediate data (JSON format and data permutation)

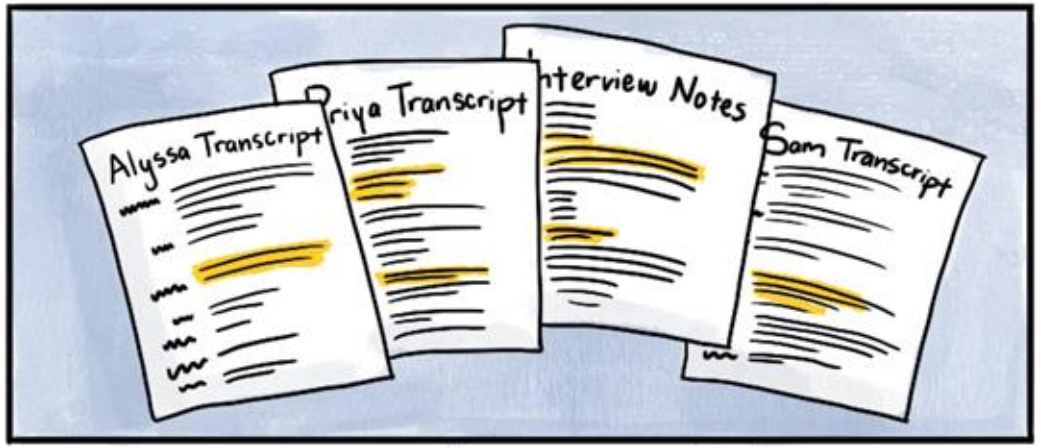
Evaluation: Cosine Similarity, Cohen's Kappa, etc



Step 1 Familiarization

LLM Prompt:

*"Please get familiar with the provided interview transcription or meeting minutes. Each interview is one-on-one. There should be only two speakers recorded. Please provide a maximum of 2 sentences **summary** for the interviewee based on the transcriptions or meeting minutes. Here is the interview from Participant {ID}{interview content}"*



https://delvetool.com/blog/thematicanalysis

GPT Response for P5 (sample):

The interviewee is a researcher who focuses on media effects from a quantitative communication perspective, particularly in sports and science. They have conducted **a project analyzing cyberbullying tactics used by former President Donald Trump on Twitter,** and they emphasize the importance of ethical considerations, data anonymization, and manual coding in their research process.

Step 2 Generating Initial Codes

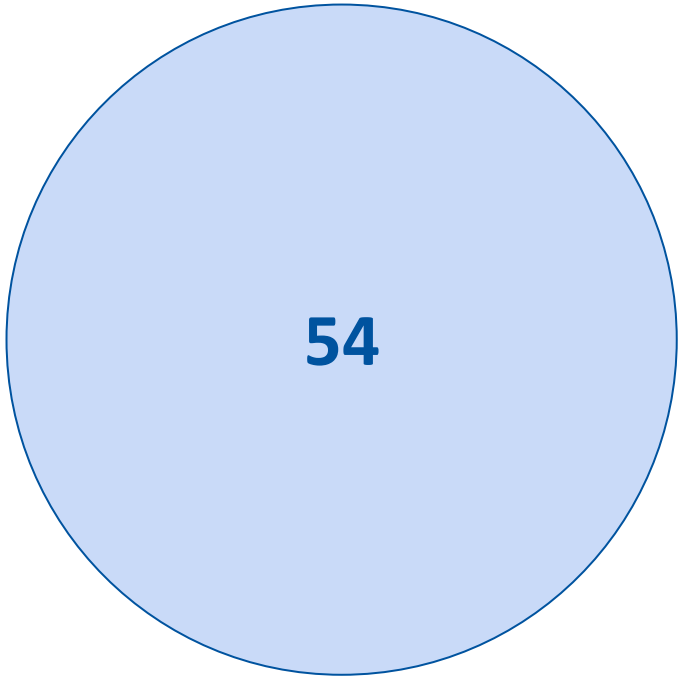
LLM Prompt:

*"I would like you to perform **qualitative coding** for an interview or meeting minute. I would like to see the results in a table. The first column is the code name. Each code should have less than 4 words. The second column is the code definition. Each **definition** should have less than 15 words. The third column is the total number of excerpts identified for that code. The fourth column displays all excerpts for that code. Do not shorten. Keep the original excerpts. You should not code the questions asked by the interviewer. You should apply codes to excerpts such as sentences or parts of sentences. For each interview, you should **identify 10-20 codes.** Please return the table as a JSON object. Here is the interview from Participant {ID} {interview content}"*



https://delvetool.com/blog/thematicanalysis

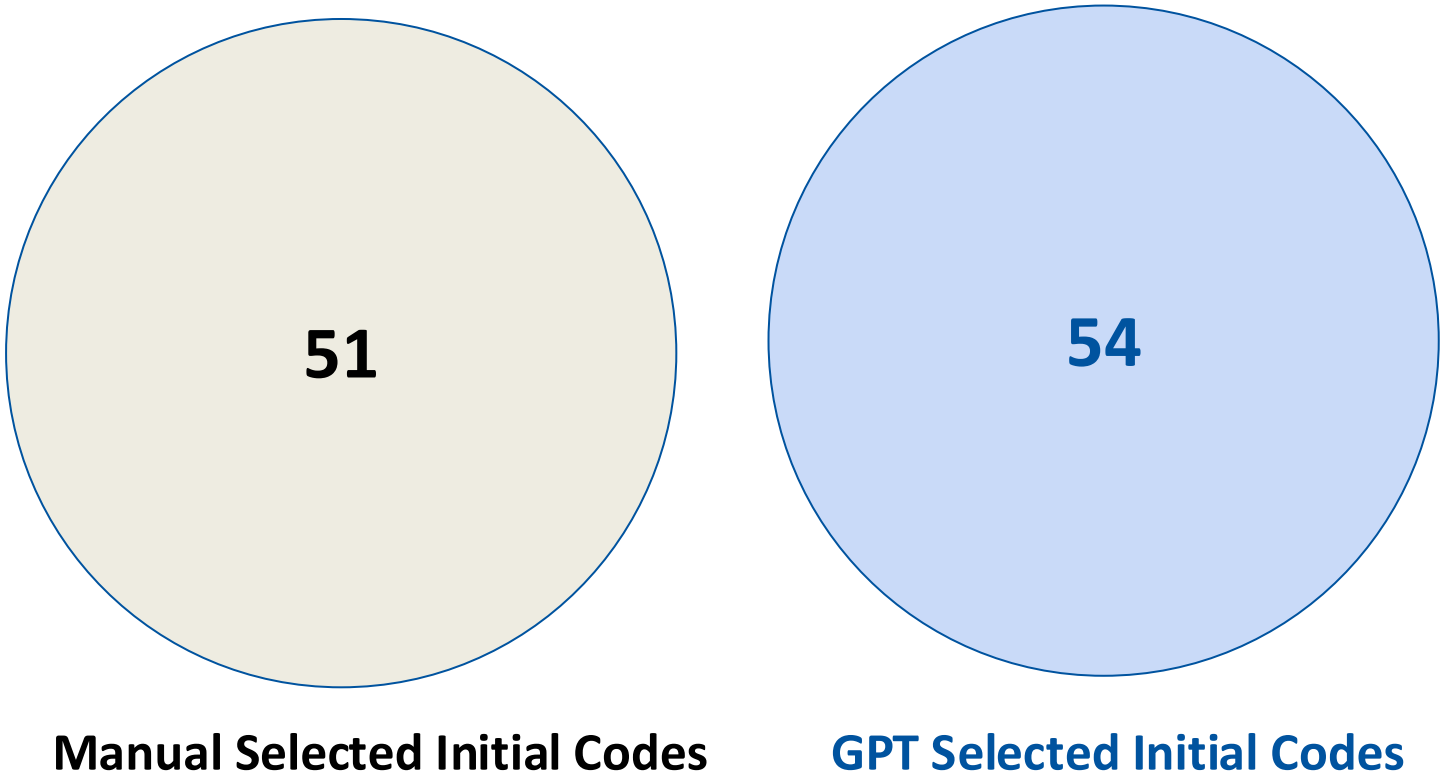
Step 2 Generating Initial Codes



GPT Selected Initial Codes

Step 2 Generating Initial Codes

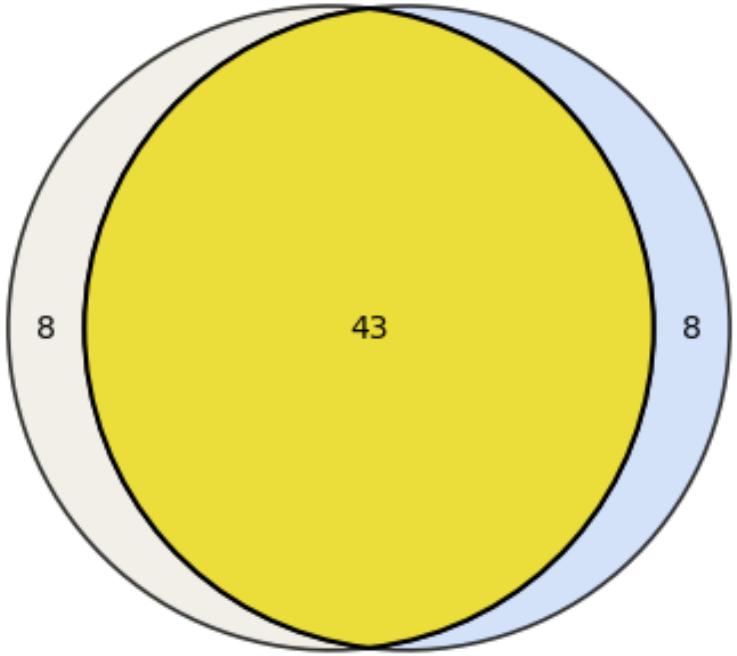
Comparison w/ manual analysis



Step 2 Generating Initial Codes

Comparison w/ manual analysis

- **85% of codes are similar**
  - **matched exactly / inferred** (i.e Role Description -> Job Description)
- 8 new code names

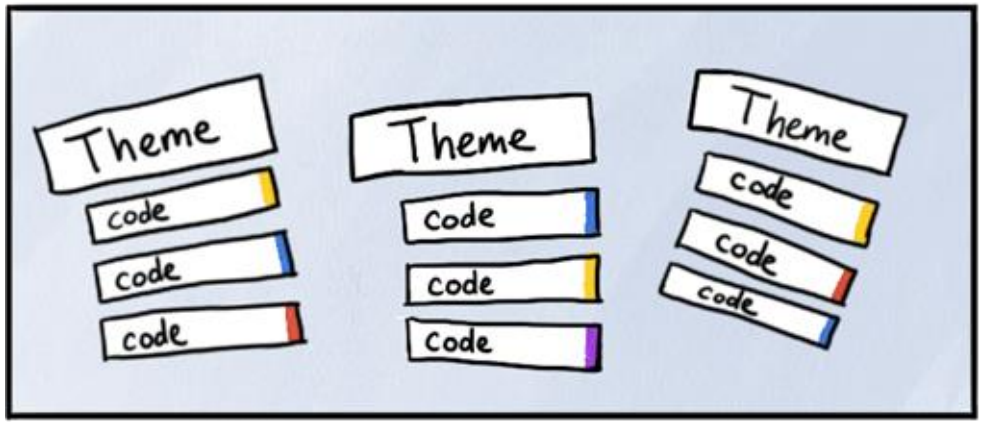


Manual Selected Initial Codes **GPT Selected Initial Codes**

Step 3 & 4 **Searching** & Review Themes

LLM Prompt:

*"I would like you now to search for **themes** as part of thematic analysis. I will give you the code name and code definition. You should determine which code can be **merged** together based on commonalities or overarching concepts. I would like to see the result in a table. The first column is the name of the theme. The second column is all codes that fit into the theme. The third column is a brief summary of the theme. Please return the table as a JSON object. Here are the codes and their definitions {response form Phase 2}.*



https://delvetool.com/blog/thematicanalysis

- Group codes into 1 theme (only name, didn't ask for definition)

Step 3 & 4 Searching & **Review** Themes

Response:

- Adjust values of ***Temperature***, regulates uncertainty/creativity
- Eg. Trial 2 remove duplicate code from 54 -> 45 and merge into 7 higher level themes

- **Capture more information**

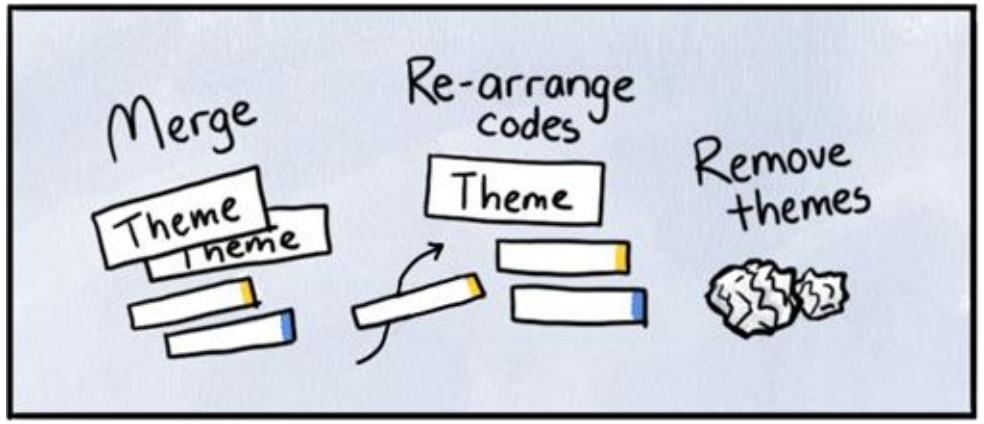
Trial 1 (*T= 0, n = 49*)	**Trial 2** (*T= 0.5, n = 45*)	**Trial 3** (*T= 1, n = 49*)	**Trial 4** (*T= 1, n = 41*)
Research Context	Research Profile	Participant Information	Role and Research Information
Data Collection	Data Collection	Data Collection	Data Collection
Data Analysis	Data Analysis	Data Management	Data Analysis
Data Quality	Data Quality and Management	Data Analysis	Data Quality
Ethical and Privacy Considerations	Ethical and Privacy Considerations	Ethical and Privacy Considerations	Ethics and Privacy
Project Management	Project Management and Improvement	Project Logistics	Challenges and Improvements
Data Management	Data Sources and Use	Future Aspects	Project Details
Data Usage		Research Tools	Data Management
Research Tools and Sources			Other Research Aspects
Special Research Areas			

Themes Identified by GPT, with multiple trials

Step 5 Defining and Naming Themes

LLM Prompt:

*"Please **review all the themes** and revise them if needed. Please provide a summary of each theme. The results should be presented in a table. The first column is the name of the theme. The second column is codes associated with the theme. **Every unique code from the original data should be included and only once**. The third column is a brief summary of the theme. Please return the table as a JSON object. Below are the themes, codes, and original summary of the themes."*



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Step 5 Defining and Naming Themes

Response:

- **10 themes** chosen by GPT
- **80%** of themes matched/associated with manual analysis themes
- **2 theme** can't be inferred (human overlooked)

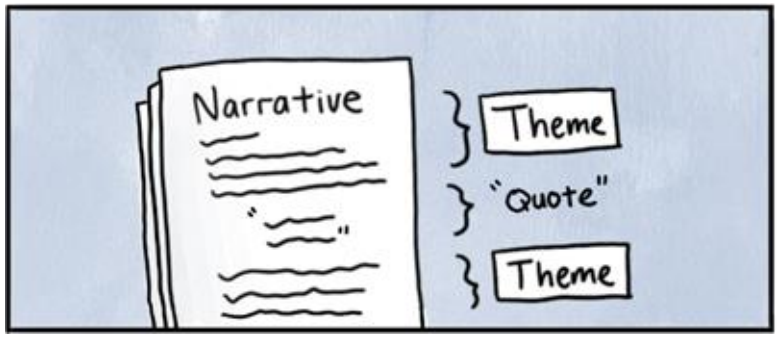
- However, **3 duplicate codes** overlap in themes. (E.g. initial code "Data Collection Tools" placed in T2, T9)

	GPT Final Themes	Associated w/ Manual Analysis
1	Data Analysis	Yes
2	Data Collection	Yes
3	Data Management	Yes
4	Data Quality	Yes
5	Data Usage	Yes
6	Ethical and Privacy Considerations	Yes
7	Project Management	Yes
8	Research Context	Yes
9	Research Tools and Sources	No
10	Special Research Areas	No

Step 6 Writing the Report

LLM Prompt:

*"As the final stage of thematic analysis, please **provide a summary and insights for the themes** based on the **excerpts.** Please keep it concise and the summary should not exceed 100 words. You may mention a few quotes with the participant ID. Below are the themes, excepts, and participant ID: {data}."*



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Step 6 Writing the Report

GPT Response

The key theme 'Ethical and Privacy Considerations' navigates around researchers' ethical hesitations dealing with the privacy issues of public Twitter users. Many respondents struggled with the idea of utilizing public information without explicit consent or ensuring anonymity; P5 remarked "just because we can scrape it doesn't mean we should." Others, like P6, noted the data was already anonymized, and P9 ensured they only used general demographic data to respect privacy.

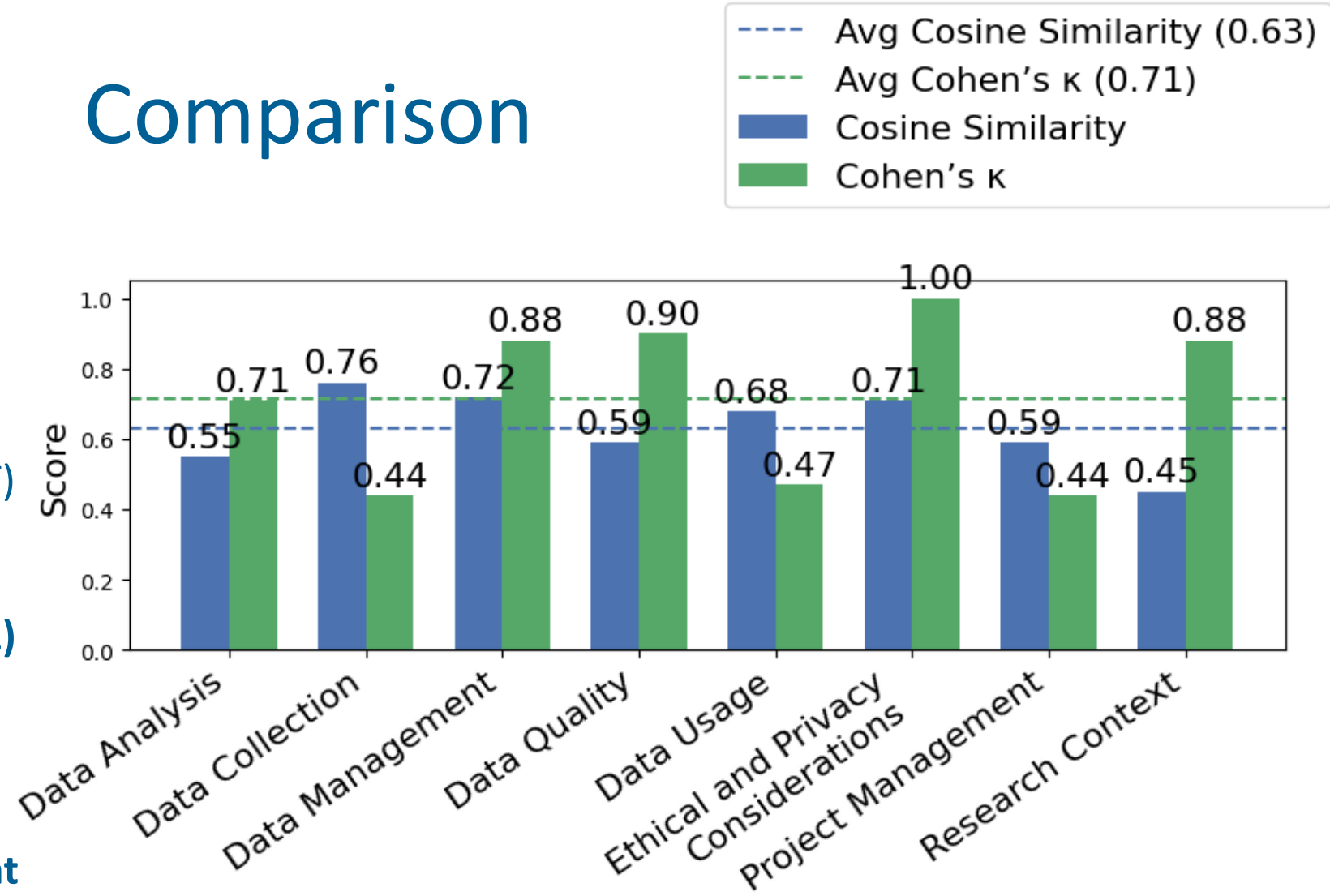
Manual Analysis

The ethical considerations show the common opinion of the participants regarding the OSN data. P1, P2, P3, and P4 delved into the right of ownership of the data published on social media platforms. While some participants (3) mentioned the importance of the IRB approval, P8 and P5 highlighted the issue of obtaining consent for collecting publicly available data "I really can't get consent from anybody because you don't know who they are, you don't know if they have like serious accounts or not." (P8). Removing identifiable information was addressed to be crucial for ensuring the anonymization of users.

Theme Comparison- "Ethical Considerations"

Comparison

- **Cosine Similarity (Avg 0.63)**
  - Semantic Textual Similarity
  - Sentence Embedding (BERT)
  - Close to 1 means identical
- **Moderate agreement (Avg 0.71)**
  - Conditions on GPT Selected Excerpts (n = 111)
  - **Human and GPT Agreement**



Further Comparison

- **Human-AI Collaboration**

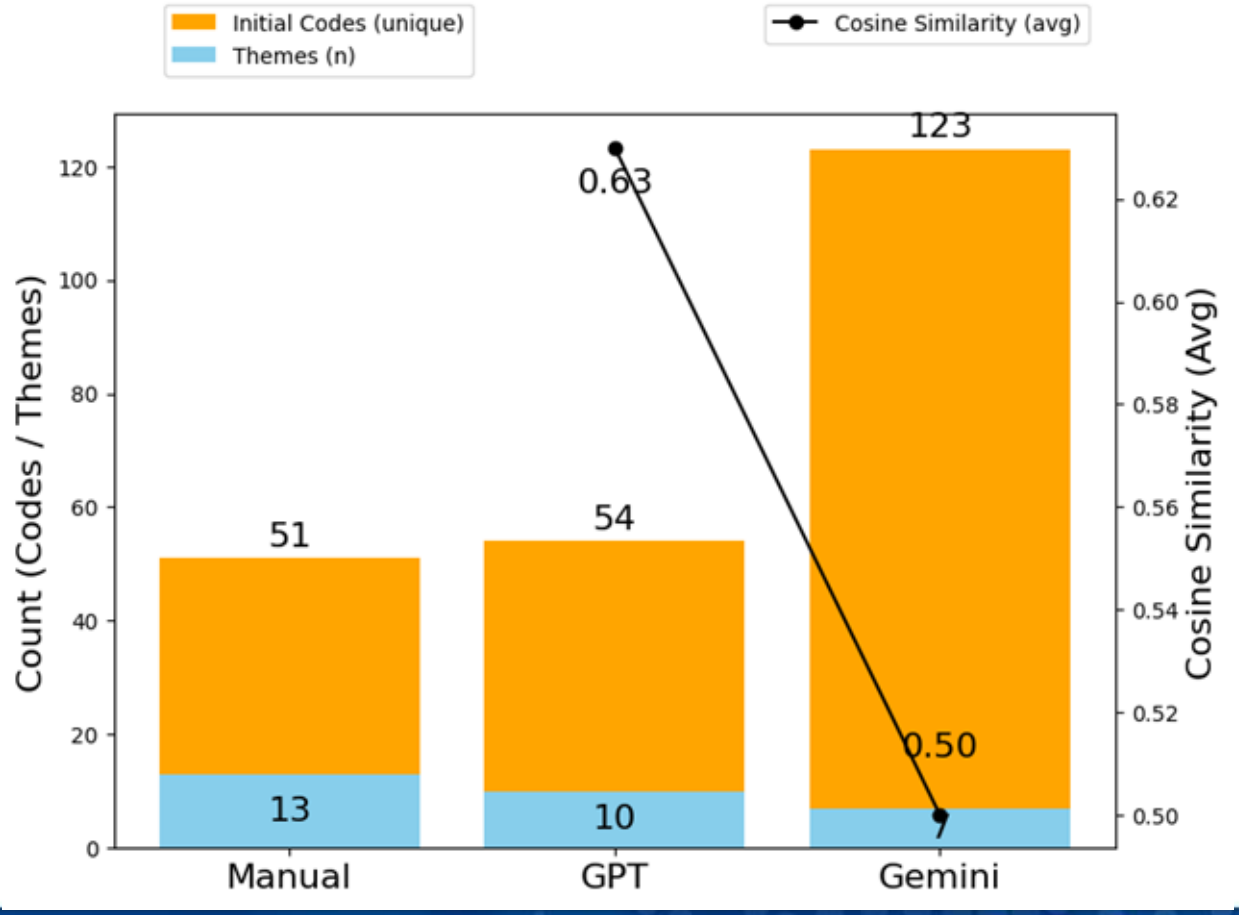
GPT slightly outperforming Gemini

- **Insight Enhancement**

LLM can uncover additional information that human experts may overlook

- **Efficiency**

Accelerates research, allowing a single researcher to effectively supervise the process.



Design Recommendation



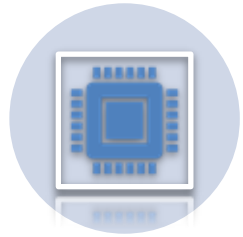
Evaluation Still Crucial



Managing Large Text Data



Efficiency with Limits



Automation Opportunities



Toward LLMs as Supportive Tools

Takeaways



from feasibility study

- **Demonstrated the feasibility** of conducting thematic analysis using LLMs with **evaluation at each step of TA**
- LLM-Assisted TA in our interview contents are **comparable and useful**



from interview content

- Primary obstacles for OSN data collection **accessibility, technical implementation, poor data quality**
- Collection Features should focus on **enhancing historical data retrieval and improving search filters**

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This publication (or program) was made possible by the National Science Foundation EPSCoR Grant No's. 2438144 and 2438145.